

CNC-204 Lathe Machine

Operation and Maintenance Manual

Model: CNC-204 | Document Version: 3.1 | Issued by: Plant Engineering Dept.

1. Introduction

The CNC-204 is a computer numerically controlled lathe designed for precision turning operations on metal and composite workpieces. It is intended for use by trained machine operators in an industrial manufacturing environment. This manual covers standard operating procedure, routine maintenance, and troubleshooting guidance for common issues.

This machine is suitable for cylindrical turning, facing, threading, and boring operations on workpieces up to 250mm in diameter and 500mm in length. Maximum spindle speed is 4500 RPM.

2. Safety Guidelines

2.1 General Safety

Operators must wear safety glasses and steel-toe footwear at all times when the machine is powered on. Loose clothing, jewelry, and long unrestrained hair are prohibited near the spindle area due to entanglement risk.

Never open the chuck guard while the spindle is rotating. The interlock system will attempt to halt spindle rotation automatically if the guard is opened, but operators should never rely on the interlock as a substitute for proper procedure.

2.2 Emergency Stop

The emergency stop button is the red mushroom-shaped button located on the front control panel, lower right side. Pressing it immediately cuts power to the spindle and all moving axes. After an emergency stop, the machine must be manually reset using the key switch before restarting.

2.3 Lockout-Tagout

Before performing any maintenance, the machine must be fully powered down and locked out using the main disconnect switch located on the rear panel. Attach a lockout tag with your name and the date before beginning work.

3. Machine Components

3.1 Control Panel

The control panel includes the main power switch, spindle start/stop buttons, jog controls for the X and Z axes, the emergency stop button, and the digital readout (DRO) screen showing current position and spindle speed.

3.2 Chuck and Spindle Assembly

The three-jaw chuck holds the workpiece and is tightened using the chuck key stored in the holder to the right of the control panel. The spindle assembly houses the drive motor and bearing system.

3.3 Tool Turret

The tool turret holds up to eight cutting tools and rotates automatically between programmed operations. Tools are secured using the turret's clamping bolts, torqued to 25 Nm.

3.4 Coolant System

The coolant reservoir is located beneath the machine bed and holds approximately 40 liters of water-soluble cutting fluid. The coolant pump activates automatically during cutting operations when programmed, or can be manually toggled using the coolant switch on the control panel.

4. Startup Procedure

1. Ensure the work area is clear and all guards are closed.
2. Turn the main power switch to ON at the rear panel.
3. Wait for the control system to complete its boot sequence (approximately 45 seconds). The DRO screen will display 'READY' when complete.
4. Perform the home reference cycle by pressing the HOME button. All axes will move to their reference position. Do not interrupt this cycle.
5. Load the workpiece into the chuck and tighten securely using the chuck key.
6. Select or load the machining program from the control panel menu.
7. Verify the tool offsets are correct before starting the cycle.

5. Normal Operation

5.1 Running a Program

Once a program is loaded and verified, press CYCLE START to begin automatic operation. The operator should remain within reach of the emergency stop button at all times during the first run of any new program.

5.2 Manual (Jog) Mode

Manual mode allows the operator to move the X and Z axes independently using the jog wheel or arrow buttons. This is typically used for tool setup, workpiece alignment, or manual measurement checks.

5.3 Spindle Speed and Feed Rate

Spindle speed can be adjusted using the override dial on the control panel, ranging from 50 percent to 120 percent of the programmed value. Feed rate override works similarly and should be reduced when cutting unfamiliar or brittle materials.

6. Routine Maintenance

6.1 Daily Checks

Before starting each shift, check coolant level, inspect the chuck jaws for wear or damage, and verify that all guards and covers are securely fastened. Wipe down exposed slideways to remove metal chips and debris.

6.2 Weekly Maintenance

Lubricate the X and Z axis guideways using the grease points located on the machine bed. Use only the specified lithium-based grease. Check the coolant filter and clean or replace if clogged.

6.3 Monthly Maintenance

Inspect the drive belt tension and replace if visible cracking or fraying is present. Check tool turret indexing accuracy using a dial indicator. Verify spindle bearing temperature does not exceed 60 degrees Celsius during extended operation.

Interval	Task	Notes
Daily	Check coolant level	Top up if below minimum line
Daily	Inspect chuck jaws	Replace if worn unevenly
Weekly	Lubricate guideways	Use lithium-based grease only
Weekly	Clean coolant filter	Replace if heavily clogged
Monthly	Check belt tension	Replace if cracked or frayed
Monthly	Check turret indexing	Use dial indicator

7. Troubleshooting

7.1 Machine Fails to Power On

Check that the main disconnect switch at the rear panel is fully in the ON position. Verify the facility power supply and check the control cabinet fuse. If the fuse is intact and power is present but the machine still fails to start, contact plant electrical maintenance.

7.2 Unusual Spindle Noise

A light grinding noise during the first few seconds after a cold start is normal and is caused by bearing lubricant redistributing. If the noise persists beyond 15 seconds or increases in volume during operation, stop the machine and inspect the spindle bearings.

7.3 Coolant Not Flowing

Check that the coolant reservoir is not empty. Inspect the coolant pump inline filter for blockages. If the pump motor is running but no coolant flows, the filter is the most likely cause.

7.4 Dimensional Inaccuracy in Finished Parts

Verify tool offsets have not shifted, and check for chuck jaw wear which can cause the workpiece to seat unevenly. Thermal expansion during long runs can also affect precision; allow the machine to reach thermal stability before machining tight-tolerance features.

7.5 Alarm Code E-104

Alarm E-104 indicates a servo following error on the Z-axis, typically caused by an obstruction or excessive cutting load. Reduce feed rate and inspect the Z-axis travel path for debris.

8. Shutdown Procedure

1. Complete or safely pause the current program.
2. Return the tool turret and axes to the home position.
3. Switch off the coolant pump and allow residual coolant to drain from the tool area.
4. Power down using the main control panel switch, then the rear disconnect switch if the machine will be idle for an extended period.
5. Clean metal chips from the bed and surrounding area before leaving the workstation.

9. Technical Specifications

Parameter	Value
Max workpiece diameter	250 mm
Max workpiece length	500 mm
Max spindle speed	4500 RPM
Spindle motor power	7.5 kW
Tool turret capacity	8 stations
X-axis travel	180 mm
Z-axis travel	520 mm
Positioning accuracy	0.005 mm
Coolant tank capacity	40 liters
Machine weight	2,400 kg
Power supply	415V, 3-phase, 50Hz
Air supply requirement	6 bar, dry and filtered

10. Tooling Guide

10.1 Recommended Tool Types

Station 1 and 2 are typically fitted with roughing turning tools for high material removal rate operations. Stations 3 and 4 are used for finishing passes requiring tighter tolerances. Stations 5 and 6 are commonly fitted with threading tools, and stations 7 and 8 are reserved for boring bars and parting tools.

10.2 Tool Life Guidelines

Carbide insert tools should be inspected every 200 parts for edge wear. A worn insert typically produces a dull or scratched surface finish and may cause dimensional drift of more than 0.02mm. Replace inserts proactively rather than waiting for visible failure, particularly on finishing operations.

10.3 Tool Offset Calibration

Tool offsets must be recalibrated whenever a tool is replaced or the turret is removed for maintenance. Use the tool setting probe located at the rear of the working envelope. Enter the offset values into the corresponding tool number in the offset table before resuming production.

11. Machine Calibration Procedure

Calibration should be performed quarterly, or immediately if dimensional inaccuracies are observed across multiple parts (see section 7.4).

1. Power down the machine and allow it to reach ambient temperature if it has been running.
2. Mount the calibration reference bar in the chuck.
3. Run the built-in calibration cycle from the maintenance menu (Menu > Maintenance > Calibration).
4. The system will automatically measure the reference bar at multiple points along its length and calculate axis compensation values.
5. Review the calibration report on the DRO screen. Deviations greater than 0.01mm across the travel range indicate a mechanical issue requiring service, not just software compensation.
6. Save the new compensation values and run a test part to confirm accuracy before returning to production.

12. Notes for Operators

The following notes are collected from operator experience and are provided as general guidance in addition to the formal procedures above.

When machining stainless steel, reduce spindle speed by approximately 20 percent compared to the programmed default for mild steel to reduce work hardening and tool wear.

If the machine has been idle overnight in a cold workshop, run a 5 minute warm-up cycle at low spindle speed before beginning precision work, to allow lubricants to reach normal operating viscosity.

Chuck jaws should be replaced as a full set, not individually, to maintain concentric gripping force on the workpiece.

13. Error Code Reference

Code	Description	Likely Cause
E-101	Spindle overload	Excessive cutting load or dull tool
E-104	Z-axis servo following error	Obstruction or excessive feed rate
E-108	X-axis servo following error	Obstruction or mechanical binding
E-112	Coolant flow fault	Empty reservoir or blocked filter
E-118	Chuck pressure low	Air supply issue or worn seals
E-122	Tool turret index fault	Misalignment or sensor fault
E-130	Spindle bearing over-temperature	Insufficient lubrication or overload

14. Glossary of Terms

DRO (Digital Readout): The screen on the control panel showing real-time axis position, spindle speed, and machine status.

Chuck: The clamping device that holds the workpiece in place during machining.

Turret: The rotating tool holder that positions different cutting tools into working position.

Feed rate: The speed at which the cutting tool moves through the workpiece material.

Tool offset: A stored correction value that accounts for the exact position and length of each individual cutting tool.

Interlock: A safety mechanism that prevents machine operation when a guard or door is open.

Lockout-tagout: A safety procedure ensuring machines are properly shut off and cannot be started during maintenance.

15. Support and Contact Information

For issues not resolved by this manual, contact the plant engineering department. When reporting an issue, please provide the machine ID, the error code if applicable, and a brief description of what was being done when the issue occurred.

Plant Engineering Helpdesk: extension 4521

Machine Manufacturer Support Line: available during business hours, contact details maintained by plant engineering department.

16. Electrical System Overview

The CNC-204 operates on a 415V three-phase supply and includes an internal step-down transformer for control circuit power at 24V DC. The main control cabinet is located on the left rear side of the machine and houses the servo drives, PLC, and safety relay module.

Access to the electrical cabinet requires lockout-tagout procedure as described in section 2.3, and should only be performed by qualified electrical maintenance personnel. The cabinet door is secured with a key-locked latch separate from the general machine access panels.

16.1 Servo Drive Indicators

Each axis servo drive has a status LED visible through the cabinet door viewing window. A solid green LED indicates normal operation. A flashing amber LED indicates a warning condition that does not stop operation but should be investigated. A solid red LED indicates a fault condition requiring immediate attention and will typically correspond to one of the error codes in section 13.

16.2 Backup Battery

The control system retains position reference data using an internal backup battery, which should be replaced every 24 months as part of scheduled maintenance. A low battery warning will appear on the DRO screen approximately 30 days before the battery is expected to fail. If the battery is not replaced in time, the machine will require a full home reference cycle on next startup, which is a normal recovery procedure.

17. Workholding and Fixturing

Standard workholding is via the three-jaw self-centering chuck described in section 3.2. For irregular or asymmetric workpieces, a four-jaw independent chuck attachment is available and should be requested from tooling stores.

When switching between chuck types, the chuck must be removed using the mounting bolts accessible through the access ports on the chuck face, and the new chuck aligned using the locating dowels before tightening. A run-out check using a dial indicator is required after any chuck change, with acceptable run-out being less than 0.02mm at the outer diameter.

For long, thin workpieces prone to deflection during turning, a tailstock center or steady rest should be used to provide additional support and prevent chatter or dimensional inconsistency along the length of the part.

18. Frequently Encountered Operator Questions

This section addresses questions commonly raised by operators during shift handovers and training sessions.

18.1 Why does the machine sometimes pause between operations?

A brief pause between programmed operations is normal and allows the tool turret to complete its indexing movement safely. This is not a fault condition. If the pause exceeds 5 seconds consistently, check for turret indexing issues as described in error code E-122.

18.2 Can the machine be left running unattended?

The CNC-204 should never be left running unattended for extended periods. Plant policy requires an operator to remain within the immediate work area whenever the spindle is rotating, in case of unexpected material failure or tool breakage.

18.3 What should be done if a part comes out oversized after a finishing pass?

First check that the correct tool offset was used and that the tool has not worn beyond its expected life (see section 10.2). If offsets and tool condition are confirmed correct, verify the machine has reached thermal stability, as bed and spindle expansion during warm-up can cause slight oversizing on early parts of a production run.

18.4 Is it normal for the coolant to appear discolored after extended use?

Mild discoloration of coolant over several weeks of use is normal due to metal fines suspended in the fluid. However, if the coolant develops a strong odor or visible separation, it should be replaced, as this indicates bacterial contamination which can affect both part finish and operator skin health.